ANDROID GENAI APP DEVELOPMENT PROJECT

Sahyadri-Samrakshane

Forest Sentinel – Citizen Science for the Western Ghats

DOCUMENT TYPE	Project Requirement Document (PRD)
DOMAIN	Environmental Technology / GenAI Mobile App
PLATFORM	Android (Native – Kotlin / Jetpack Compose)
VERSION	1.0 | Draft for Review
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1. Executive Summary

Sahyadri-Samrakshane (Kannada: *Sahyadri Protection*) is a native Android application designed as a citizen-science tool for real-time ecological alert reporting across the Western Ghats biodiversity hotspot. The app empowers trekkers, local villagers, and forest watchers to instantly report forest fires, landslides, illegal logging, and notable wildlife sightings — complete with high-precision GPS coordinates and photographic evidence — directly to the forest department's central monitoring dashboard.

Powered by GenAI capabilities, Firebase real-time sync, and CameraX imaging, the app is built to work reliably even in low-connectivity mountainous terrain through offline-first caching. It represents a convergence of community policing, disaster early-warning, and environmental stewardship.

2. Problem Statement

The Core Challenge

The Western Ghats — stretching across Kerala, Karnataka, Goa, Maharashtra, Tamil Nadu, and Gujarat — are one of the world's eight 'Hottest Hotspots' of biological diversity and serve as the primary water tower for peninsular India. Despite their ecological importance, these regions face severe and ongoing threats:

■ Forest Fires	Seasonal and man-made fires destroy thousands of hectares annually. By the time fire reports travel through informal channels to the forest department, the damage is irreversible.

■ Illegal Logging	Timber poaching and illegal tree cutting are rampant in remote zones. Lack of direct reporting mechanisms allows offenders to escape.
■■ Landslides	The Ghats experience frequent landslides during monsoon. Early community detection can save lives, but no standardised alert channel exists.
■ Location Ambiguity	Even when locals manage to report an incident, conveying precise location in dense forest terrain is extremely difficult without GPS coordinates.
■■ Response Lag	The delay between community observation and official action — due to absence of a direct digital bridge — often renders the response ineffective.

3. Vision & Objectives

Vision Statement

"To create an always-on, community-powered digital shield for the Western Ghats — where every trekker becomes a forest guard, and every alert becomes a seed of ecological protection."

Strategic Objectives

O1	**Real-Time Reporting**	Enable instant ecological alert submission with photo + GPS within 60 seconds of incident detection.
O2	**Offline Resilience**	Ensure zero data loss in no-signal zones by caching reports locally and auto-syncing on reconnection.
O3	**Precise Location**	Provide GPS coordinates accurate to ±5 metres using FusedLocationProviderClient for rapid field response.
O4	**Community Education**	Deliver contextual eco-guidelines and safe behaviour tips within the app.
O5	**Dashboard Integration**	Sync all alerts to a Firebase-backed dashboard for forest department officials to monitor and respond.

4. Stakeholders & Target Users

Primary Users	Trekkers & hikers in the Western Ghats	High	Report alerts, view status, read eco-tips
Primary Users	Local villagers & farmers near forest areas	High	Report alerts with minimal tech friction
Secondary Users	Forest department officers & guards	High	Receive, verify, and act on alerts via dashboard

Secondary Users	NGOs & environmental researchers	Medium	Analyse alert data for patterns and studies
Tertiary Users	Disaster management authorities	Medium	Respond to landslide & fire alerts
Developers	Student developers (academic project)	High	Build, maintain, and demonstrate the app

5. Functional Requirements

FR-01	**Alert Submission**	Must	User can submit an Ecological Alert including: Alert Type, Photo, GPS coordinates, optional description, and timestamp.
FR-02	**Alert Type Selection**	Must	App presents four alert categories: Forest Fire, Landslide, Illegal Tree Cutting, Wildlife Sighting. Each has a distinct icon and colour code.
FR-03	**CameraX Integration**	Must	In-app camera powered by CameraX library captures high-resolution photos (minimum 4MP). Photo is auto-attached to the active alert report.
FR-04	**GPS Capture**	Must	FusedLocationProviderClient obtains GPS lock. Latitude and longitude are displayed on-screen during reporting and stored with the alert.
FR-05	**Offline Caching**	Must	If network is unavailable, the alert is stored in local Room database. A background WorkManager job retries upload every 15 minutes until successful.
FR-06	**Alert Status Tracking**	Must	Each submitted alert displays one of three statuses: Reported, Verified, Team Dispatched. Status updates are pushed via Firebase real-time listener.
FR-07	Education Module	Should	A dedicated section provides eco-tips: safe trekking behaviour, what to do during a forest fire or wildlife encounter, guidelines for eco-sensitive zones.
FR-08	Alert History	Should	Users can view their previously submitted alerts with status, photo thumbnail, GPS location, and submission timestamp.
FR-09	GenAI Alert Assist	Should	Gemini API integration to (a) auto-suggest alert type from photo, and (b) generate a structured description of the scene to assist reporting.
FR-10	Map View	Could	Integration with Google Maps SDK to display reported alerts as pins on a local map within 50 km radius.
FR-11	Push Notifications	Could	FCM-powered push notifications alert users when their reported incident status changes from Reported to Verified or Team Dispatched.
FR-12	Multilingual Support	Could	App UI available in English and Kannada to maximise accessibility for local Karnataka communities.

6. User Flow & App Screens

Screen 1	**Splash / Onboarding**	App logo on forest green background. Brief 3-slide onboarding explaining the mission. Permission requests: Camera, Location, Notification.
Screen 2	**Home Dashboard**	Greeting with user's region. Quick-action button: '+ Report Alert' (amber CTA). Stats panel: My Reports, Local Alerts. Navigation: Home | Report | History | Learn.
Screen 3	**Alert Type Selection**	Grid of 4 cards: Forest Fire (red-orange icon), Landslide (brown icon), Illegal Tree Cutting (dark green icon), Wildlife Sighting (teal icon). Each card tappable with haptic feedback.
Screen 4	**Capture & Report**	CameraX viewfinder (full-screen). GPS coordinate overlay at bottom. Capture button. After capture: Review photo, add optional text note. 'Submit Alert' button.
Screen 5	**Submission Confirmation**	Success animation (leaf icon). Alert ID displayed. Estimated review time. Option to 'Report Another' or 'View My Alerts'. Offline banner if cached.
Screen 6	**My Alerts / History**	List of past alerts sorted by recency. Status chip (colour-coded). Tap to expand: photo, GPS, description, status timeline.
Screen 7	**Education / Eco-Tips**	Card-based tips: Fire Safety, Encountering Wildlife, Landslide Precautions, Leave No Trace. Each card expandable with detailed guidance.
Screen 8	**Settings**	Profile, Notification preferences, Language toggle (EN / KN), About & Acknowledgements.

7. Technical Implementation

Architecture Overview

The application follows the MVVM (Model-View-ViewModel) architecture pattern with a Repository layer, ensuring clean separation of concerns, testability, and scalability.

UI Layer	Jetpack Compose	Declarative UI with Nature-Green theme tokens. Composable screens for each user flow. StateFlow-driven reactivity.
ViewModel Layer	Kotlin Coroutines + Flow	Business logic isolation. Handles GPS updates, image capture state, and alert submission lifecycle.
Repository Layer	Repository Pattern	Single source of truth. Decides whether to fetch from Room (offline) or Firebase (online).
Local Storage	Room Database	Stores unsynced alerts with columns: alertId, type, photoPath, latitude, longitude, description, syncStatus, timestamp.
Remote Sync	Firebase Firestore	Real-time NoSQL database. Alerts collection synced bi-directionally. Status updates pushed to device.
Background Sync	WorkManager	PeriodicWorkRequest checks network and uploads cached Room alerts to Firestore. Guaranteed execution even post-restart.
Camera	CameraX (Jetpack)	ImageCapture use-case for high-res photo. Preview use-case for viewfinder. LifecycleCameraController for simplified setup.
Location	FusedLocationProviderClient	HIGH_ACCURACY priority. Real-time updates during capture. Fallback to last known location if fix unavailable.
GenAI	Gemini 1.5 Flash API	Multimodal prompt: send captured image → receive alert type suggestion + auto-generated incident description.
Notifications	Firebase Cloud Messaging	Server-side trigger on Firestore document update → FCM push to user device on status change.
Maps	Google Maps SDK / OSMdroid	Display alert location pins. Offline tile caching via OSMdroid for no-signal areas.
Auth	Firebase Anonymous Auth	Lightweight anonymous auth to associate alerts with a device identity without requiring sign-up.

Data Model — Alert Document (Firestore)

Field	Type	Values	Description
alertId	String	UUID	Auto-generated unique identifier
userId	String	Device ID	Firebase Anonymous Auth UID
alertType	Enum	FOREST_FIRE | LANDSLIDE | ILLEGAL_LOGGING | WILDLIFE	Category of ecological event
latitude	Double	±5m accuracy	GPS latitude from FusedLocationProvider
longitude	Double	±5m accuracy	GPS longitude from FusedLocationProvider
photoUrl	String	Firebase Storage URL	Uploaded photo reference
description	String	Optional (GenAI-generated or manual)	Scene description
status	Enum	REPORTED | VERIFIED | TEAM_DISPATCHED	Current response status
timestamp	Timestamp	ISO 8601	Alert creation time (UTC)
syncStatus	Enum	PENDING | SYNCED	Local Room sync state (offline support)

8. Non-Functional Requirements

Performance	NFR-01	Alert submission (photo + GPS + upload) must complete within 8 seconds on a stable 4G connection.
Performance	NFR-02	App cold-start time must not exceed 3 seconds on a mid-range Android device (2GB RAM).
Reliability	NFR-03	Offline-cached alerts must not be lost across app restarts, device reboots, or OS-level process kills.
Reliability	NFR-04	GPS coordinate acquisition must succeed within 30 seconds in open terrain; fallback to last-known location otherwise.
Usability	NFR-05	UI text minimum font size 14sp. WCAG AA contrast ratio (4.5:1) for all text on green backgrounds.
Usability	NFR-06	First-time user should be able to submit an alert without any training within 2 minutes of installing.
Security	NFR-07	No personally identifiable information collected without explicit consent. Anonymous auth only.
Security	NFR-08	All network communication over HTTPS / TLS 1.3. Firebase Security Rules restrict write access to authenticated users.
Maintainability	NFR-09	Code coverage minimum 60% for ViewModel and Repository layers (JUnit + Mockito).
Compatibility	NFR-10	Target SDK 34 (Android 14). Minimum SDK 26 (Android 8.0 Oreo).

9. Impact Goals

■ Environmental Security	The Western Ghats supply water to 245 million people across six Indian states. Early detection and reporting of deforestation and fires directly preserves this critical watershed.
■■ Disaster Management	Landslides in the Ghats kill dozens annually. Citizen-reported early warnings — transmitted with GPS precision — can enable pre-emptive evacuations and save lives.
■ Community Policing	Engaging local villages as active stakeholders in forest protection fosters ownership, reduces illegal activity through social deterrence, and strengthens the human-nature relationship.

■ Education & Awareness	In-app eco-tips cultivate a generation of informed eco-citizens who understand biodiversity conservation, responsible trekking, and wildlife coexistence.
■ Data-Driven Conservation	Aggregated, geotagged alert data provides researchers and policymakers with evidence-based insights into threat hotspots, seasonal patterns, and enforcement gaps.

10. Success Criteria (Student Evaluation)

SC-01	Critical	**Offline-First Reporting**	App caches alert locally when network is unavailable. WorkManager automatically syncs report to Firebase within 15 minutes of connectivity restoration. Zero data loss verified by test.
SC-02	Critical	**Live GPS Display**	Latitude and longitude coordinates are visible on-screen in the Report screen during photo capture, updating in real time with minimum 1 update per 5 seconds.
SC-03	Critical	**Nature-Friendly UI**	All screens use the defined green/earth-tone colour palette. Primary: #1B4332, Secondary: #52B788, Accent: #E9C46A. No grey/blue corporate palette.
SC-04	Critical	**Four Alert Types**	All four alert categories (Forest Fire, Landslide, Illegal Tree Cutting, Wildlife Sighting) are selectable with distinct icons, colours, and Firebase storage paths.
SC-05	Important	**Status Tracking**	Submitted alerts display status (Reported / Verified / Team Dispatched). Status changes made in Firebase console are reflected in app within 5 seconds via real-time listener.
SC-06	Important	**CameraX Integration**	Photos are captured via CameraX (not Intent/ACTION_IMAGE_CAPTURE). Resolution minimum 1920x1080. Photo stored in Firebase Storage and URL written to Firestore document.
SC-07	Important	**GenAI Feature**	At least one GenAI-powered feature is implemented and functional (e.g., Gemini-based alert type suggestion from photo, or auto-description generation).
SC-08	Bonus	**Multilingual (EN/KN)**	App fully supports Kannada language toggle with all UI strings translated using Android string resources.
SC-09	Bonus	**Map View**	Recent alerts displayed as map pins in a Google Maps or OSMdroid fragment with correct GPS coordinates.

11. Constraints & Assumptions

Technical Constraints

- The app targets Android only (no iOS deliverable required for this academic submission).
- Firebase free-tier (Spark Plan) is assumed; Blaze Plan may be needed for Firebase Storage beyond 1GB.
- Gemini API access requires a valid Google AI Studio API key — students must provision their own key.
- CameraX requires a physical Android device for full testing; emulator camera may behave differently.
- FusedLocationProvider requires Google Play Services; AOSP devices without Play Services are out of scope.
- Offline sync via WorkManager is constrained to minimum 15-minute intervals per Android OS battery restrictions.

Assumptions

- Forest department officials will access alerts through the Firebase console or a separate web dashboard (not this app).
- Users grant Camera and Location (ACCESS_FINE_LOCATION) permissions on first launch.
- The app is used in Karnataka / Western Ghats region; time zone is IST (UTC+5:30).
- Alert photos will be under 5MB per submission (JPEG compression applied before upload).
- Students will mock the 'Verified' and 'Team Dispatched' status updates manually via Firebase console for demonstration.
- Connectivity restoration is detected via ConnectivityManager.NetworkCallback API.

12. Appendix – Technology Stack Summary

Language	Kotlin 1.9+	Primary development language
UI Framework	Jetpack Compose	Declarative, reactive UI
Architecture	MVVM + Repository	Clean architecture pattern
Camera	CameraX (androidx.camera)	High-res capture, preview, lifecycle-aware
Location	FusedLocationProviderClient	High-accuracy GPS, battery-efficient
Local DB	Room (androidx.room)	Offline alert caching, SQLite abstraction
Remote DB	Firebase Firestore	Real-time sync, NoSQL cloud database
File Storage	Firebase Storage	Photo upload and CDN delivery
Auth	Firebase Auth (Anonymous)	Device identity without sign-up friction
Notifications	Firebase Cloud Messaging	Status-change push notifications
Background Jobs	WorkManager	Guaranteed offline sync retry

GenAI	Gemini 1.5 Flash (Google AI)	Photo analysis, alert description generation
Maps	Google Maps SDK / OSMdroid	Alert visualisation (optional offline tiles)
DI	Hilt (Dagger)	Dependency injection
Async	Kotlin Coroutines + Flow	Reactive data streams, async operations
Testing	JUnit 4 + Mockito + Espresso	Unit and integration tests
CI/CD	GitHub Actions (optional)	Automated build and test pipeline
Design	Material 3 (You)	Adaptive colour, accessible components
Colour Scheme	#1B4332 / #52B788 / #E9C46A	Forest dark / leaf green / amber accent
Min SDK	API 26 (Android 8.0)	Broad device coverage
Target SDK	API 34 (Android 14)	Latest platform features

■ Sahyadri-Samrakshane — Project Requirement Document

This document is prepared for academic review as part of the Android GenAI App Development module. All Firebase projects, API keys, and cloud resources are to be provisioned by the student team. Status: Draft v1.0 — subject to revision based on faculty feedback.